

STM32F446RE GPIO Pin Map — Reconfigurable BMS

Board: NUCLEO-F446RE (Nucleo-64) **Status:** LOCKED — verify in CubeMX before wiring **Last updated:** M1 Step 1

Conflicts Resolved

Issue	Original	Fix
PA2/PA3 USART2 conflict	PA2 → Q3 GPIO, PA3 → ADC tap 1	Moved Q1–Q3 GPIO to PB4/PB5/PB6. PA3 freed (but still reserved for USART2)
PA8/PA9 not ADC pins	PA8 → "ADC2_IN?", PA9 → "ADC2_IN?"	PA8 → I2C3_SCL (INA260 B). NTCs moved to PC2/PC3/PC5/PA6
Q4 load disconnect missing	No pin assigned	PB10 assigned
INA260 address conflict	2× INA260 on one I2C bus, no A0/A1 pads	Split to I2C1 (String A) + I2C3 (String B)

GPIO Outputs — Switch Control

Pin	Function	Connection	Connector	Pull-down
PB4	GPIO_Output Q1 NPN base (String A main)		Arduino D5	100kΩ to GND
PB5	GPIO_Output Q2 NPN base (String B main)		Arduino D4	100kΩ to GND
PB6	GPIO_Output Q3 NPN base (Precharge)		Arduino D10	100kΩ to GND
PB10	GPIO_Output Q4 NPN base (Load disconnect)		Arduino D6	100kΩ to GND

Safety: All four have 100kΩ base pull-downs. Default state at power-up/reset = all MOSFETs OFF. **Drive circuit per switch:** STM32 pin → 1kΩ → 2N2222A base. 100kΩ pull-down on base. Collector → 10kΩ → IRF4905 gate (tied to source = battery+). Emitter → GND.

ADC Inputs — Voltage Taps (6 channels)

All on **ADC1**, scanned via DMA.

Pin	ADC Channel	Signal	Divider	Max Vin	Max Vadc	Connector
PA0	ADC1_IN0	String A Tap 1 (cell A1+)	33k/10k	~4.2V	~0.98V	Arduino A0
PA1	ADC1_IN1	String A Tap 2 (cell A1+ A2+)	33k/10k	~8.4V	~1.95V	Arduino A1
PA4	ADC1_IN4	String A Tap 3 (top of string)	33k/10k	~12.6V	~2.93V	Arduino A2
PB0	ADC1_IN8	String B Tap 1 (cell B1+)	33k/10k	~4.2V	~0.98V	Arduino A3
PB1	ADC1_IN9	String B Tap 2 (cell B1+ B2+)	33k/10k	~8.4V	~1.95V	Morpho CN10-24
PC4	ADC1_IN14	String B Tap 3 (top of string)	33k/10k	~12.6V	~2.93V	Morpho CN10-34

Divider ratio: $10k / (33k + 10k) = 0.2326$ **Cell voltage computed in firmware:** $V_{cell1} = V_{tap1}$, $V_{cell2} = V_{tap2} - V_{tap1}$, $V_{cell3} = V_{tap3} - V_{tap2}$ **Resolution:** 12-bit ADC, 3.3V ref $\rightarrow 0.8mV/step \rightarrow \sim 3.4mV$ per step at battery side **Worst-case subtraction error:** $\sim 5mV$ (adequate for 50–100mV fault detection)

ADC Inputs — NTC Thermistors (6 channels)

All on **ADC1**, same DMA scan as voltage taps.

Pin	ADC Channel	Signal	Connector
PC0	ADC1_IN10	NTC Cell A1	Arduino A5
PC1	ADC1_IN11	NTC Cell A2	Arduino A4
PC2	ADC1_IN12	NTC Cell A3	Morpho CN7-35
PC3	ADC1_IN13	NTC Cell B1	Morpho CN7-37
PC5	ADC1_IN15	NTC Cell B2	Morpho CN10-6
PA6	ADC1_IN6	NTC Cell B3	Arduino D12

Circuit: $10k\Omega$ NTC + $10k\Omega$ pull-up to 3.3V (divider). At 25°C $\rightarrow \sim 1.65V$. Lookup table in firmware.

I2C — Current Sensors (INA260)

Pin	Function	Connection	Connector
PB8	I2C1_SCL	INA260 String A	Arduino D15
PB9	I2C1_SDA	INA260 String A	Arduino D14
PA8	I2C3_SCL	INA260 String B	Arduino D7
PC9	I2C3_SDA	INA260 String B	Morpho CN10-1

Why 2 buses: INA260 modules in hand have no A0/A1 address pads. Same default address → can't share bus. **INA260 specs:** Integrated 2mΩ shunt, ±15A range, bus voltage 0–36V. No external shunt needed.

USART — Debug & Data Logging

Pin	Function	Connection	Connector
PA2	USART2_TX	ST-Link Virtual COM Port	Arduino D1 (DO NOT USE)
PA3	USART2_RX	ST-Link Virtual COM Port	Arduino D0 (DO NOT USE)

Hardwired on Nucleo via solder bridges SB13/SB14 (default ON). Non-negotiable. **Data format:** CSV at 10Hz over USB-UART. 14 channels: 6 voltages, 2 currents, 6 temperatures + timestamp + topology state.

CAN Bus — Optional (stretch goal)

Pin	Function	Connection	Connector
PA11	CAN1_RX	SN65HVD230 transceiver	Morpho CN10-14
PA12	CAN1_TX	SN65HVD230 transceiver	Morpho CN10-12
Not needed for M1–M3. Wire only if ahead of schedule.			

Reserved / Special Pins

Pin	Purpose	Notes
PA5	LD2 (Green user LED)	Keep for heartbeat / debug blink. Arduino D13
PA13	SWDIO	ST-Link debug — do not use as GPIO
PA14	SWCLK	ST-Link debug — do not use as GPIO
PC13	B1 USER button	Available as emergency-stop input if needed

Spare Pins

Pin	Has ADC?	Connector	Potential Use
PA7	Yes (IN7)	Arduino D11	Spare ADC — bus voltage monitor, extra sensor
PA5	Yes (IN5)	Arduino D13	Spare ADC if LED not needed
PA9	No	Arduino D8	Spare GPIO
PA10	No	Arduino D2	Spare GPIO
PB3	No	Arduino D3	Spare GPIO (caution: SWO trace pin)
PB7	No	Morpho CN7-21	Spare GPIO
PC7	No	Arduino D9	Spare GPIO
PC6	No	Morpho CN10-4	Spare GPIO
PC8	No	Morpho CN10-2	Spare GPIO

ADC1 DMA Scan Order

ADC scans in ascending channel order. DMA buffer layout:

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Index	Channel	Pin	Signal
0]	IN0	PA0	Vtap A1
1]	IN1	PA1	Vtap A2
2]	IN4	PA4	Vtap A3
3]	IN6	PA6	NTC Cell B3 ← NTC interleaved by channel number
4]	IN8	PB0	Vtap B1
5]	IN9	PB1	Vtap B2
6]	IN10	PC0	NTC Cell A1
7]	IN11	PC1	NTC Cell A2
8]	IN12	PC2	NTC Cell A3
9]	IN13	PC3	NTC Cell B1
10]	IN14	PC4	Vtap B3
11]	IN15	PC5	NTC Cell B2

Voltage taps and NTCs are interleaved in the raw buffer. Use named constants in firmware, NOT raw indices.

Morpho-Only Pins Summary

5 ADC pins + 1 I2C pin are only on morpho connectors (not Arduino headers):

Pin	Morpho Location	Signal
PB1	CN10 pin 24	Vtap B2
PC4	CN10 pin 34	Vtap B3
PC2	CN7 pin 35	NTC Cell A3
PC3	CN7 pin 37	NTC Cell B1
PC5	CN10 pin 6	NTC Cell B2
PC9	CN10 pin 1	I2C3_SDA (INA260 B)

Use jumper wires from these morpho pins to your zero PCB.

Pin Count Summary

Category	Pins Used
GPIO outputs (switches)	4
ADC inputs (voltage taps)	6
ADC inputs (NTC)	6
I2C (current sensors)	4
USART (debug)	2
CAN (optional)	2
Reserved (LED, SWD, button)	4
Total allocated	28
Spare (with ADC)	2
Spare (GPIO only)	7+

CubeMX Verification Checklist

- ☐ PA2 → USART2_TX, PA3 → USART2_RX (default)
- ☐ PB4, PB5, PB6, PB10 → GPIO_Output (push-pull, no pull-up/down, low speed)
- ☐ All 12 ADC pins → ADC1_INx (verify channel numbers match this document)
- ☐ ADC1 → Scan mode ON, DMA continuous, 12-bit resolution
- ☐ PB8 → I2C1_SCL, PB9 → I2C1_SDA
- ☐ PA8 → I2C3_SCL, PC9 → I2C3_SDA
- ☐ PA5 → GPIO_Output (LD2)

- ☐ Confirm PA8 has NO ADC option (validates our earlier finding)
- ☐ Confirm PA9 has NO ADC option (validates our earlier finding)
- ☐ No yellow conflict triangles in CubeMX pin view
- ☐ Generate code, verify it compiles clean